

Auteur: Stan Van Campenhout

Studierichting: Informatica

Oefeningenreeks 4A nr. 48.4

$$\begin{aligned}\int \ln\left(\frac{x^2}{x^2-1}\right) dx &= x \ln\left(\frac{x^2}{x^2-1}\right) - \int x d\left(\ln\left(\frac{x^2}{x^2-1}\right)\right) \\&= x \ln\left(\frac{x^2}{x^2-1}\right) - \int x \cdot \frac{x^2-1}{x^2} \cdot \frac{(x^2-1)2x - x^2 \cdot 2x}{(x^2-1)^2} dx \\&= x \ln\left(\frac{x^2}{x^2-1}\right) - \int \frac{-2}{x^2-1} dx = x \ln\left(\frac{x^2}{x^2-1}\right) + 2 \int \frac{1}{(x-1)(x+1)} dx \\&\rightarrow \frac{1}{(x-1)(x+1)} = \frac{A}{x-1} + \frac{B}{x+1} \\&\Rightarrow A = \frac{1}{1+1} = \frac{1}{2} \\&\Rightarrow B = \frac{1}{-1-1} = -\frac{1}{2} \\&= x \ln\left(\frac{x^2}{x^2-1}\right) + \int \frac{1}{x-1} dx - \int \frac{1}{x+1} dx \\&= x \ln\left(\frac{x^2}{x^2-1}\right) + \ln|x-1| - \ln|x+1| + c\end{aligned}$$

References